Story Board

Figure A1. Homepage and Accessible Heritage-Story Navigation

Figure A1 presents the visitor's entry into the platform. It shows how the homepage introduces the museum through chaptered visual storytelling and how visitors can use narration, language, subtitle, text-size, and navigation controls to access the platform's principal modules.

Figure A2. Visitor Registration, Email Verification, and Sign-In

Figure A2 illustrates how a visitor creates and accesses a platform account. The sequence covers account registration, email ownership verification, profile completion, and secure sign-in before the visitor uses account-based functions such as bookmarks, submissions, bookings, and saved progress.

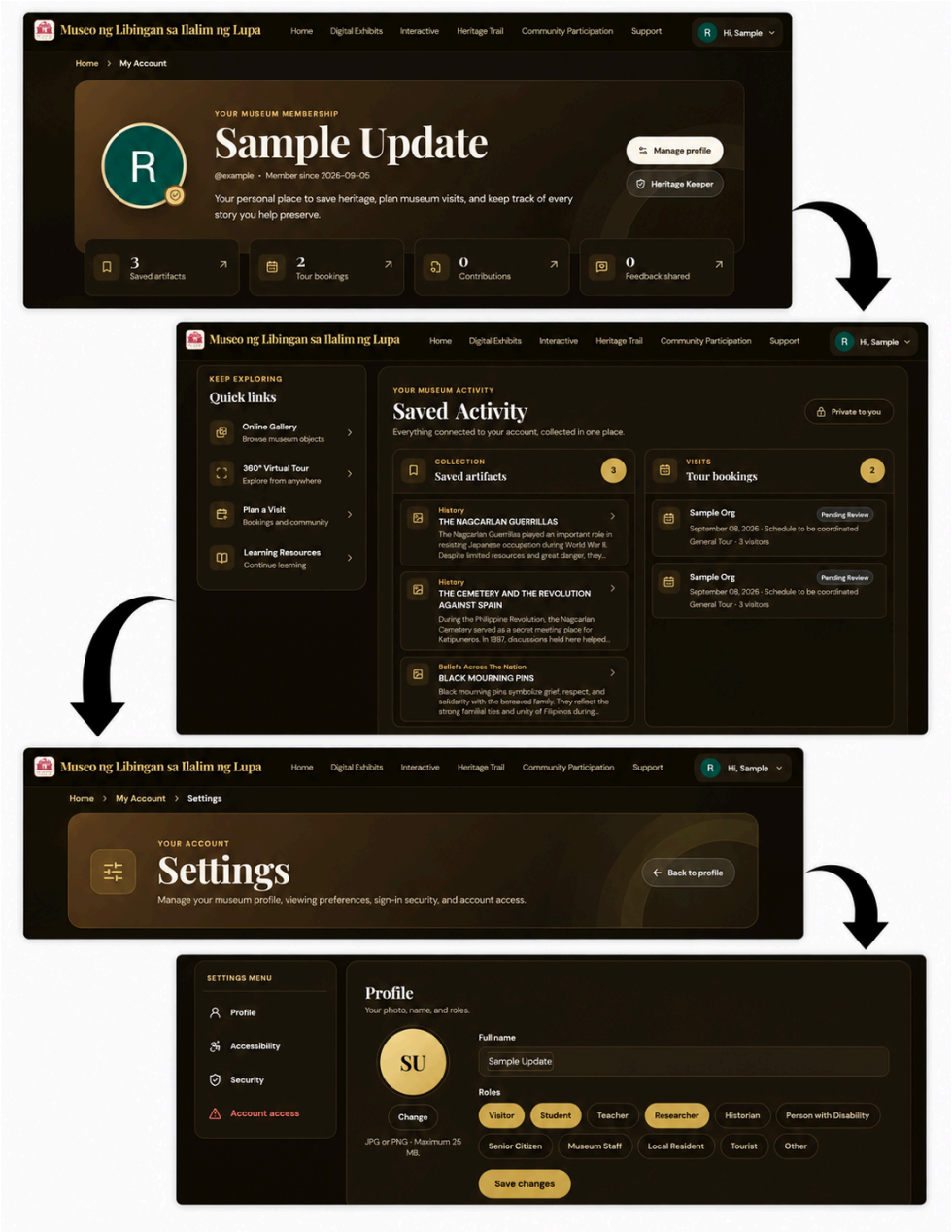

Figure A3. Visitor Account, Saved Content, and Preference Management

Figure A3 shows the registered visitor's personal account workspace. It allows the user to review profile information and account activity, return to saved museum content, monitor available participation records, and manage language, theme, text, profile, and security preferences.

Figure A4. Digital Exhibits Portal, Online Gallery, and Educational Resource Discovery and Access

Figure A4 presents the platform's Digital Exhibits portal and its digital heritage and learning services. Visitors may select the Online Gallery for artifact exploration, use search and filters, open detailed artifact records, inspect available visual or 3D media, and save artifacts to their account. Visitors may also access the Educational Resources library, use search, filtering, sorting, and view controls, open selected resource details, and use available view or download options.

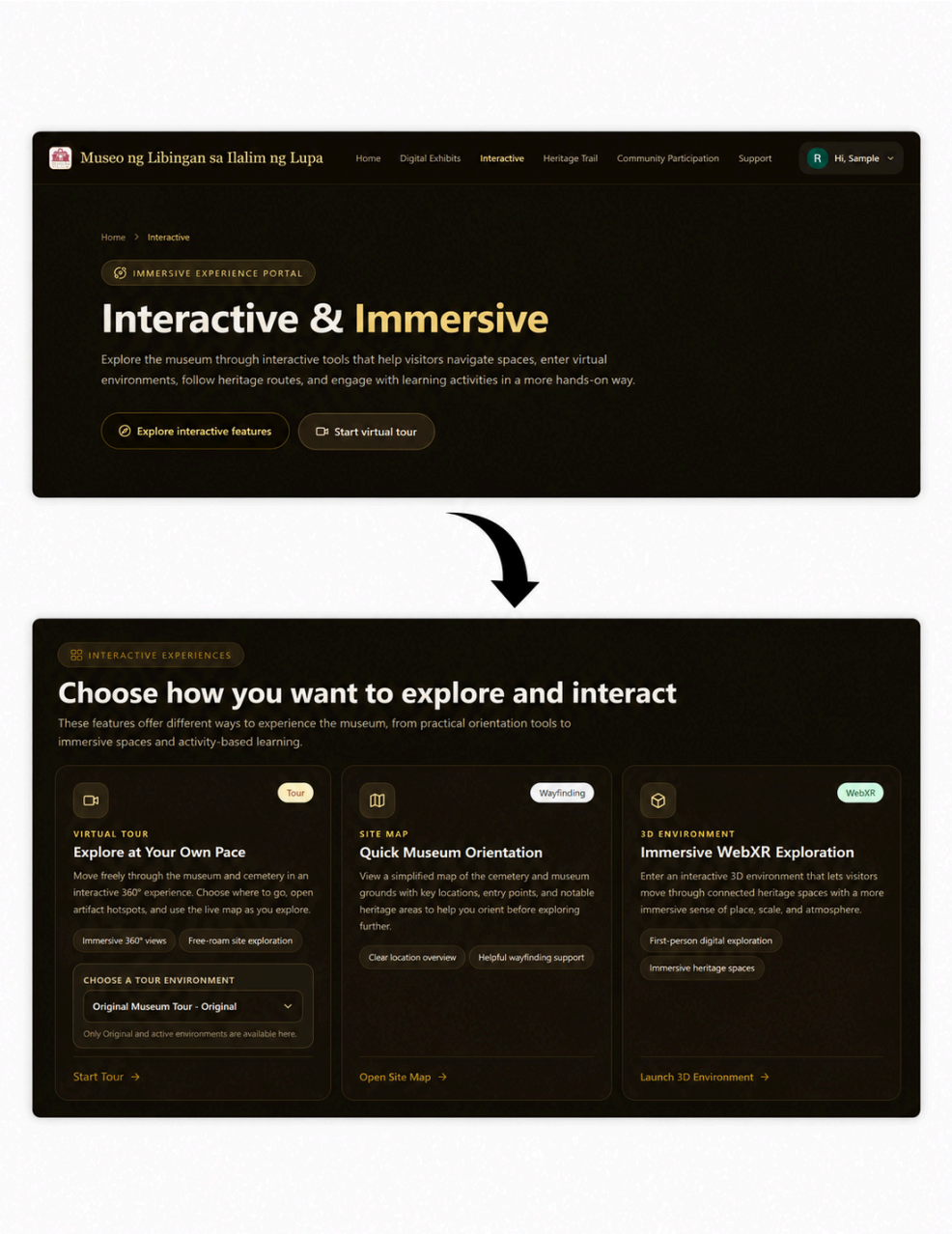

Figure A5. Interactive and Immersive Experience Selection

Figure A5 shows how visitors choose among the platform's major interactive experiences. From the Interactive and Immersive portal, the user may begin a selected 360-degree Virtual Tour, open the Site Map for orientation, or launch the WebXR-enabled 3D Environment.

Figure A6. 360-Degree Virtual Tour Navigation and Artifact Interaction

Figure A6 illustrates the free-roaming Virtual Tour of the cemetery and museum. Visitors can look around panoramic scenes, move between locations through navigation hotspots, use the live minimap for orientation, open artifact or information hotspots, and use available narration, language, subtitles, gyroscope, or VR-mode controls when supported.

Figure A7. Interactive Site Map and Aerial-View Navigation

Figure A7 presents the platform's spatial-orientation tools. Visitors can inspect the museum grounds through the aerial site map, select location markers to view their corresponding

information panel, zoom or pan the map, and switch between the illustrated aerial map and the interactive 360-degree aerial panorama.

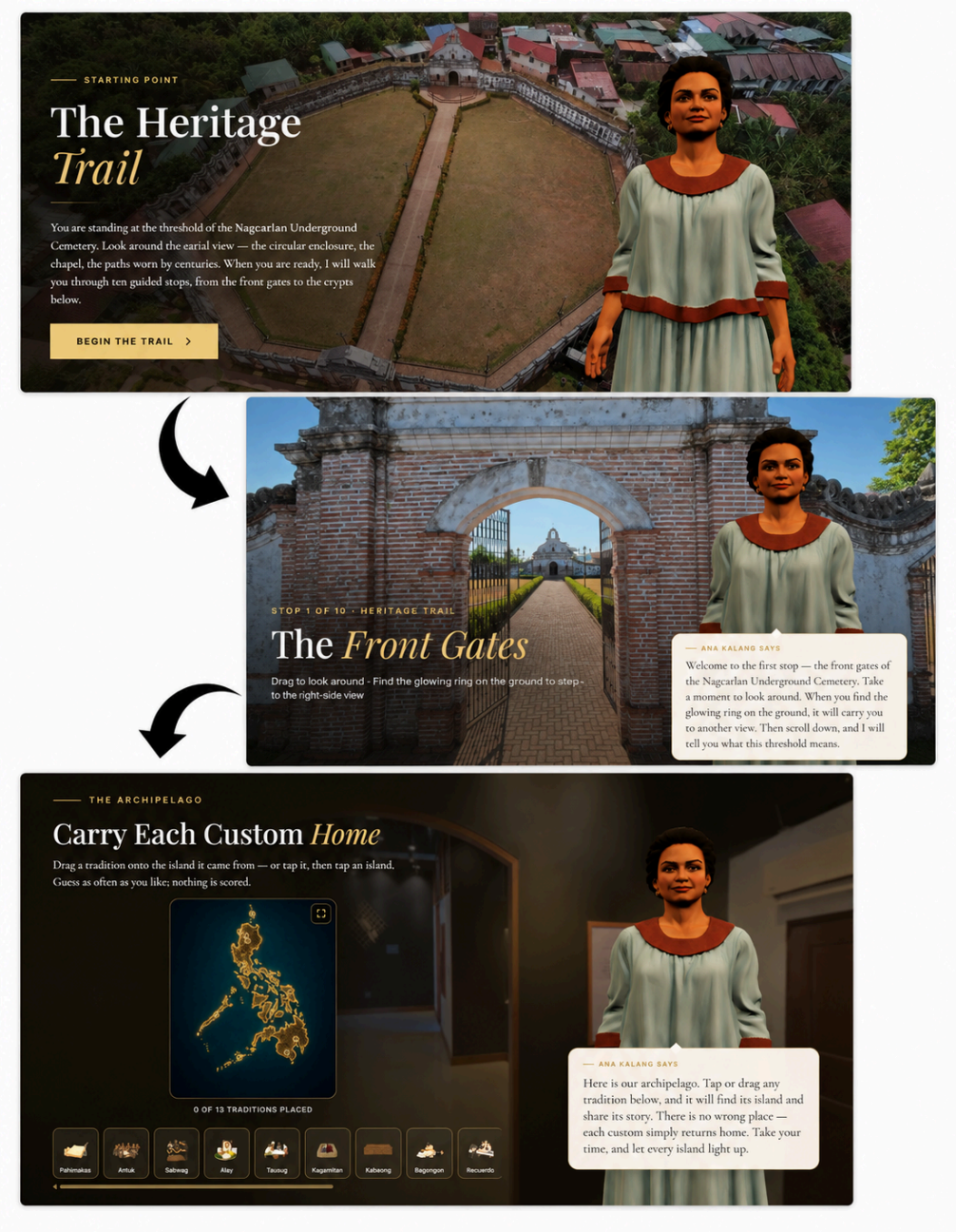

Figure A8. Guided Heritage Trail, Media, and Learning Activities

Figure A8 shows the visitor's progression through the ten-stop Heritage Trail. Each stop combines 360-degree scenes, bilingual narration, historical media, and self-check activities, while completed stops contribute to the visitor's Heritage Keeper progress and earned-fragment state.

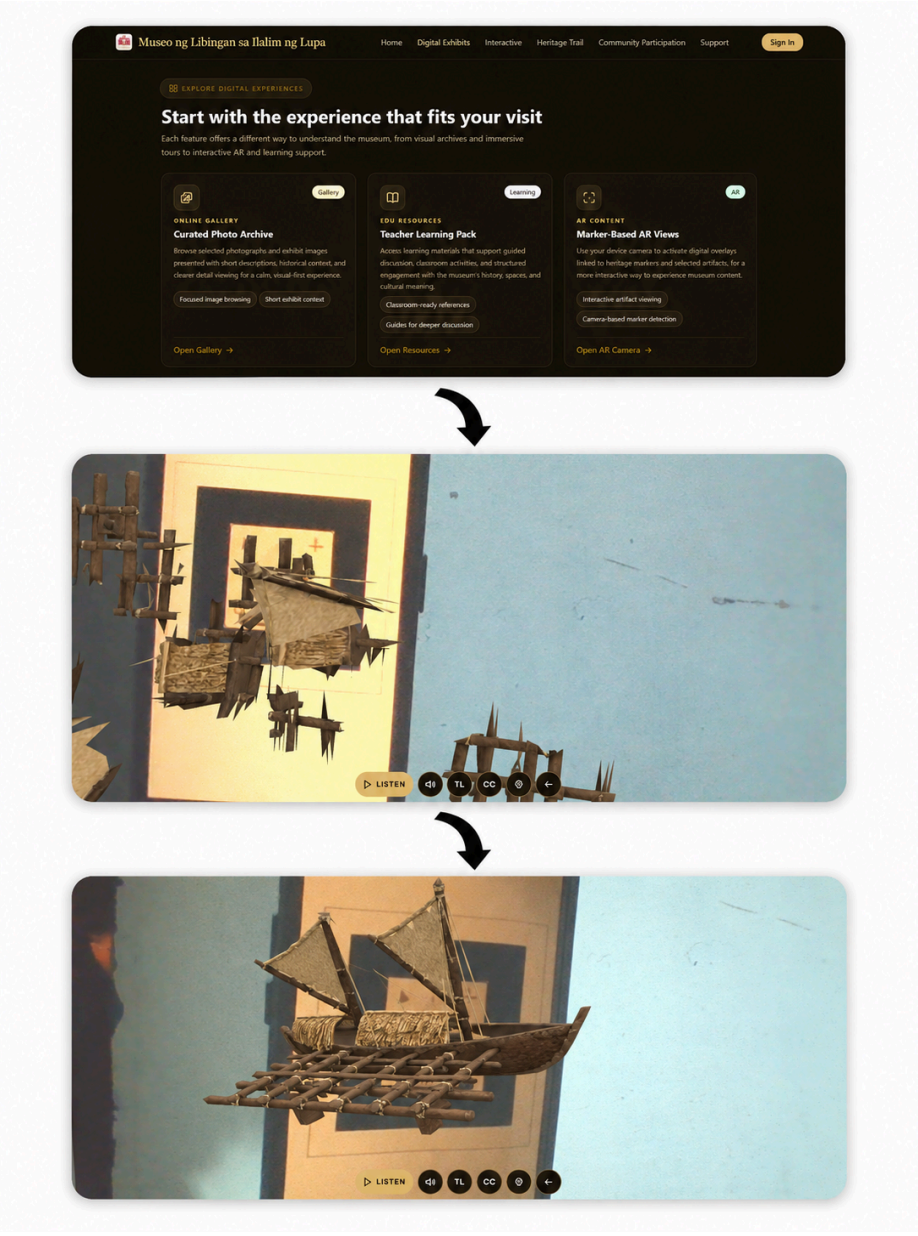

Figure A9. AR Artifact Detection and Interactive Restoration

Figure A9 illustrates the marker-based AR experience. From AR Content Detection, the system requests camera access, detects a registered marker, displays the fragmented 3D artifact with alignment guidance, restores the artifact after the correct viewing angle is maintained, and provides the View in Your Space action or supported device AR viewer.

Figure A10. WebXR 3D Environment Navigation and Interaction

Figure A10 presents the browser-based 3D museum environment. Visitors can select and navigate the Church, Museum, or Underground environment using desktop, touch, or on-screen movement controls, interact with designated objects and hotspots, review the Explorer Record or player-information panel, and enter WebVR or WebAR on compatible devices when evaluated.

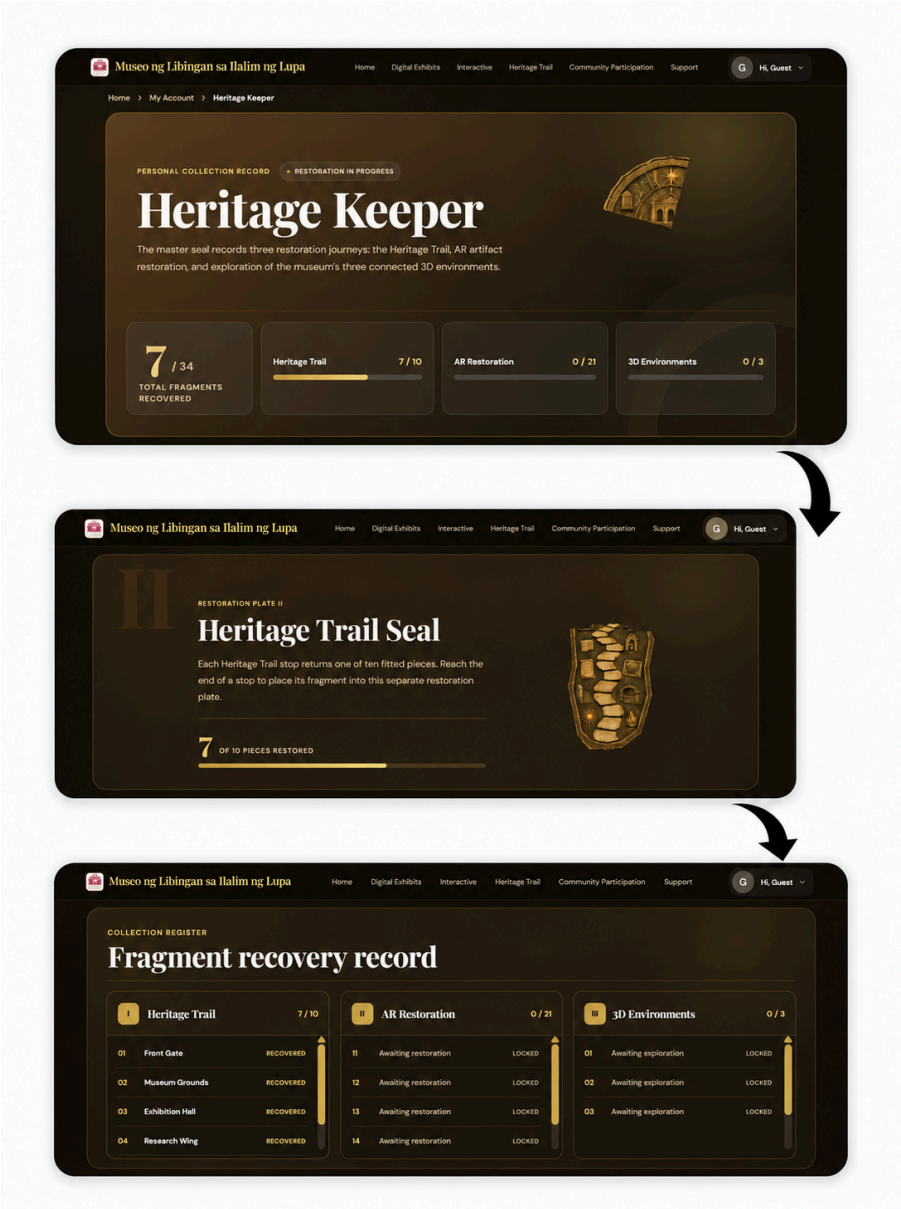

Figure A11. Heritage Keeper Progress and Digital Seal Restoration

Figure A11 shows how a registered visitor's activity is recorded across the Heritage Trail, AR restoration, and WebXR environments. Recovered pieces appear in the AR Restoration Seal, Heritage Trail Seal, 3D Environment Seal, and fragment-recovery ledger, while completion of the required experiences contributes to restoration of the master heritage seal.

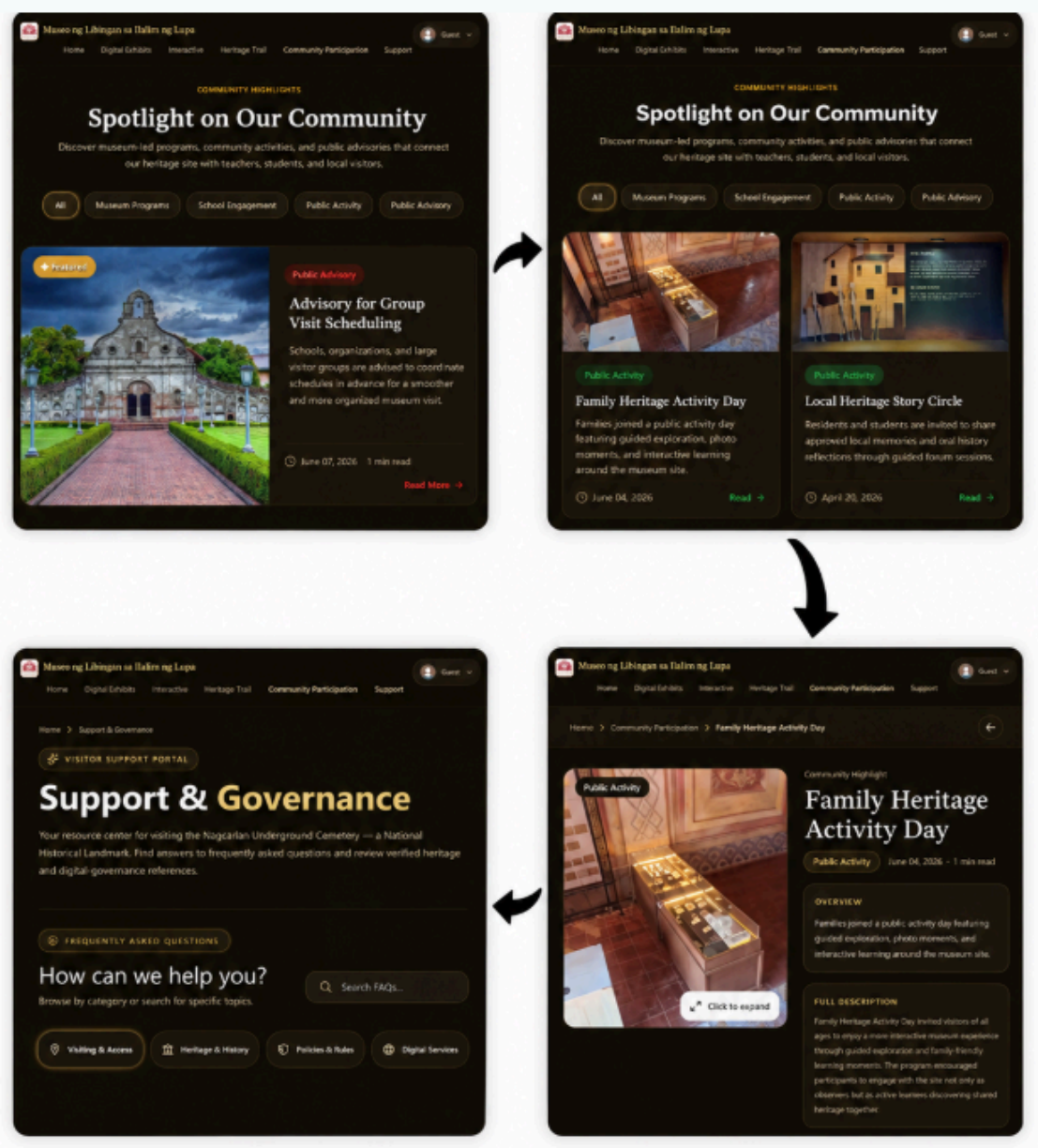

Figure A12. Community Highlights

Figure A12 shows the Community Highlights feature of the developed platform, where users can browse published community-related content and filter entries according to available categories such as museum programs, school engagement, public activity, and public advocacy. These functions were designed to organize community-oriented content, encourage public engagement, and provide users with additional channels for accessing relevant museum information and support services.

Figure A13. E-Participation

Figure A13 presents the E-Participation feature of the developed platform, which provides users with several channels for contributing information and participating in museum-related activities. From the main E-Participation interface, users may access the Visitor

Feedback, Teacher Corner, and Stories Submit functions, each intended for a specific type of contribution.

Figure A14. Booking Calendar and Feedback Submission

Figure A14 presents the visitor-facing Community Highlights, Support, guided-visit booking, Visitor Feedback, and E-Participation workflows. Visitors may browse museum

programs, school-engagement posts, public activities, or advisories, consult the Support page, submit a Request a Booking, monitor booking status, provide Visitor Feedback, contribute materials through the Teacher Corner, or submit personal and cultural narratives through Share My Story.

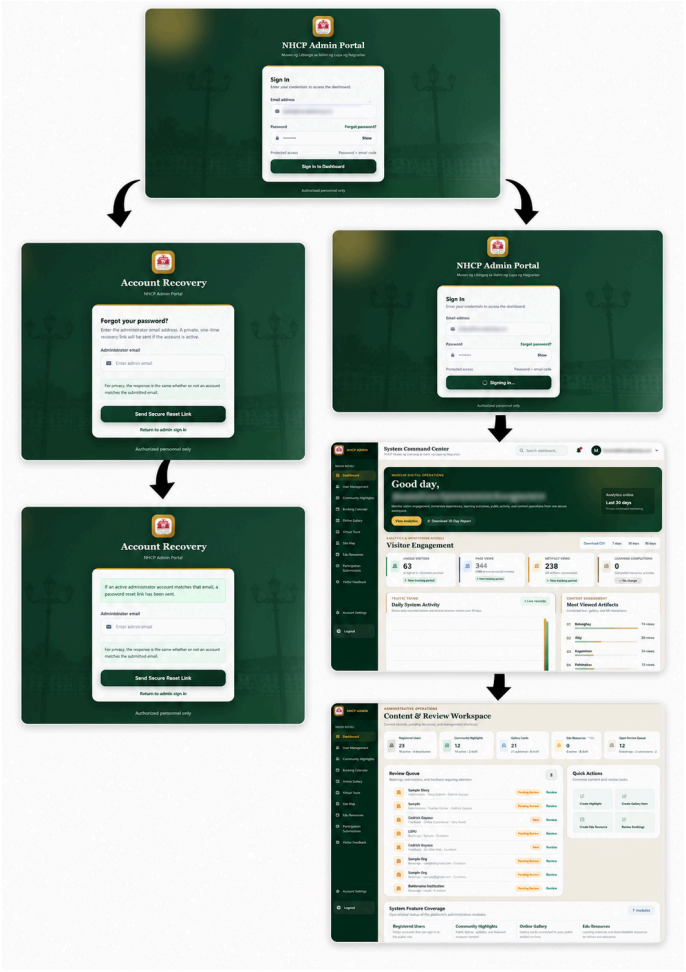

Figure A15. Administrator Authentication, Password Recovery, Dashboard, Analytics, and Review Workspace

Figure A15 shows how authorized museum personnel access the protected NHCP Admin Portal through credential-based authentication and use the Forgot Password, recovery-email, and Reset Password process when necessary. After successful login, the administrator dashboard

presents registered-user, content, and review-queue summaries together with Visitor Engagement, Most Viewed Artifacts, Virtual Tour Statistics, Device Usage, Activity Results, Recent User Activity, Engagement by Module, and the Content and Review Workspace with Review Queue and Quick Actions.

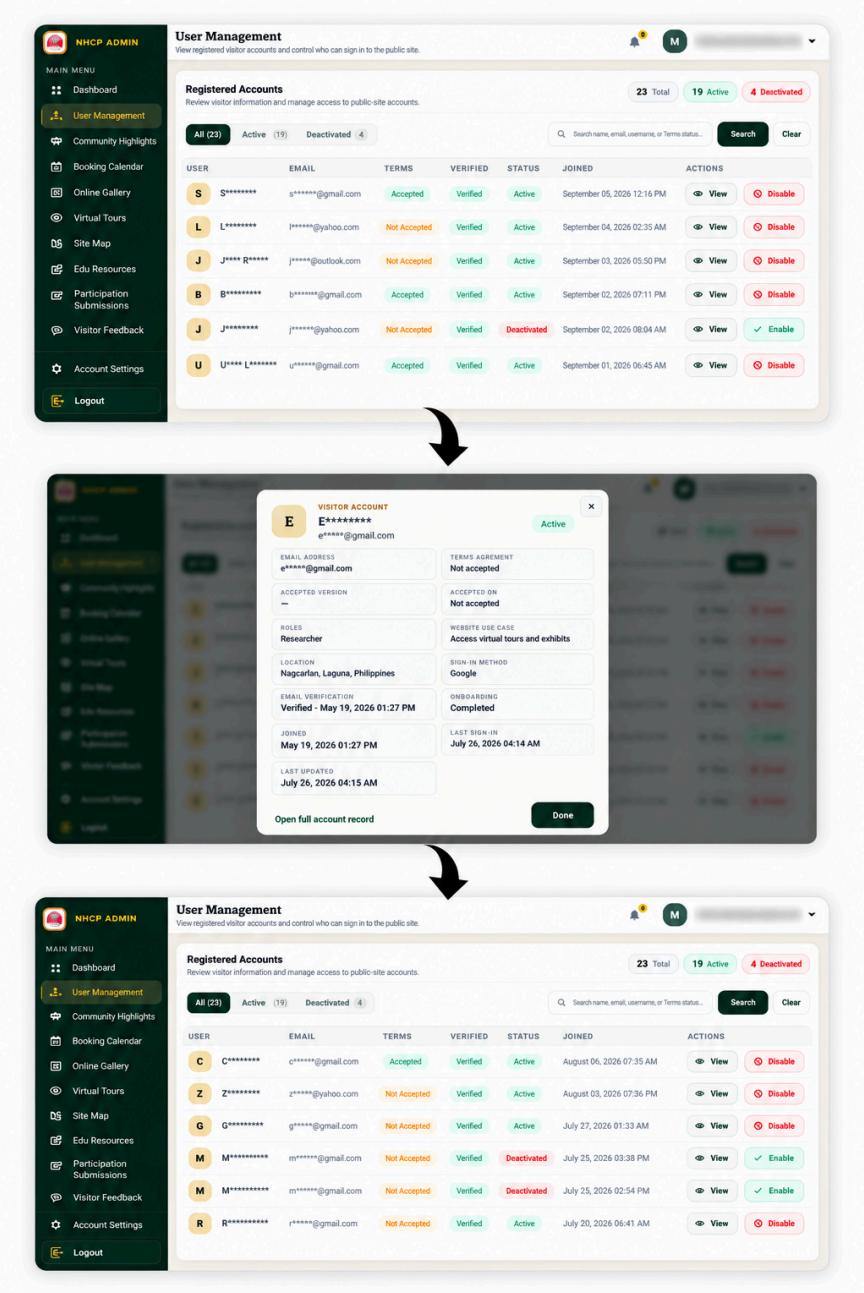

Figure A16. Public-User Account Review and Access Management

Figure A16 shows how an administrator uses User Management to locate and review registered visitor accounts. Authorized personnel can use search controls, inspect an account quick-view or full detail panel and activity summaries, and perform Deactivate or Activate actions to manage public access.

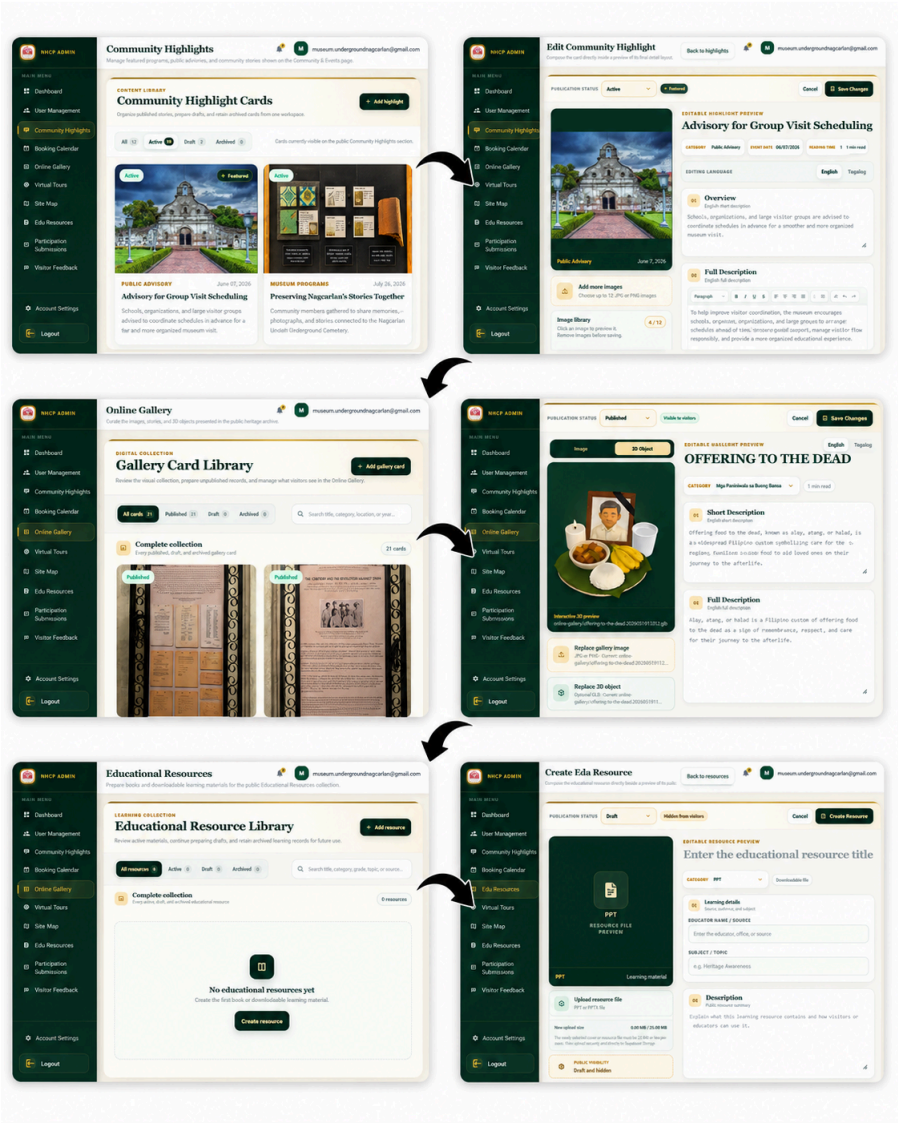

Figure A17. Community Highlight, Online Gallery, and Educational Resource Content Management

Figure A17 presents the content-management workflows for Community Highlight Cards, Gallery Cards, and the Educational Resource Library. Administrators can create or edit content, manage media and descriptive information, review saved records, use Publish or Feature actions, manage Active, Draft, or Archived status, and verify that published content appears in the appropriate public module.

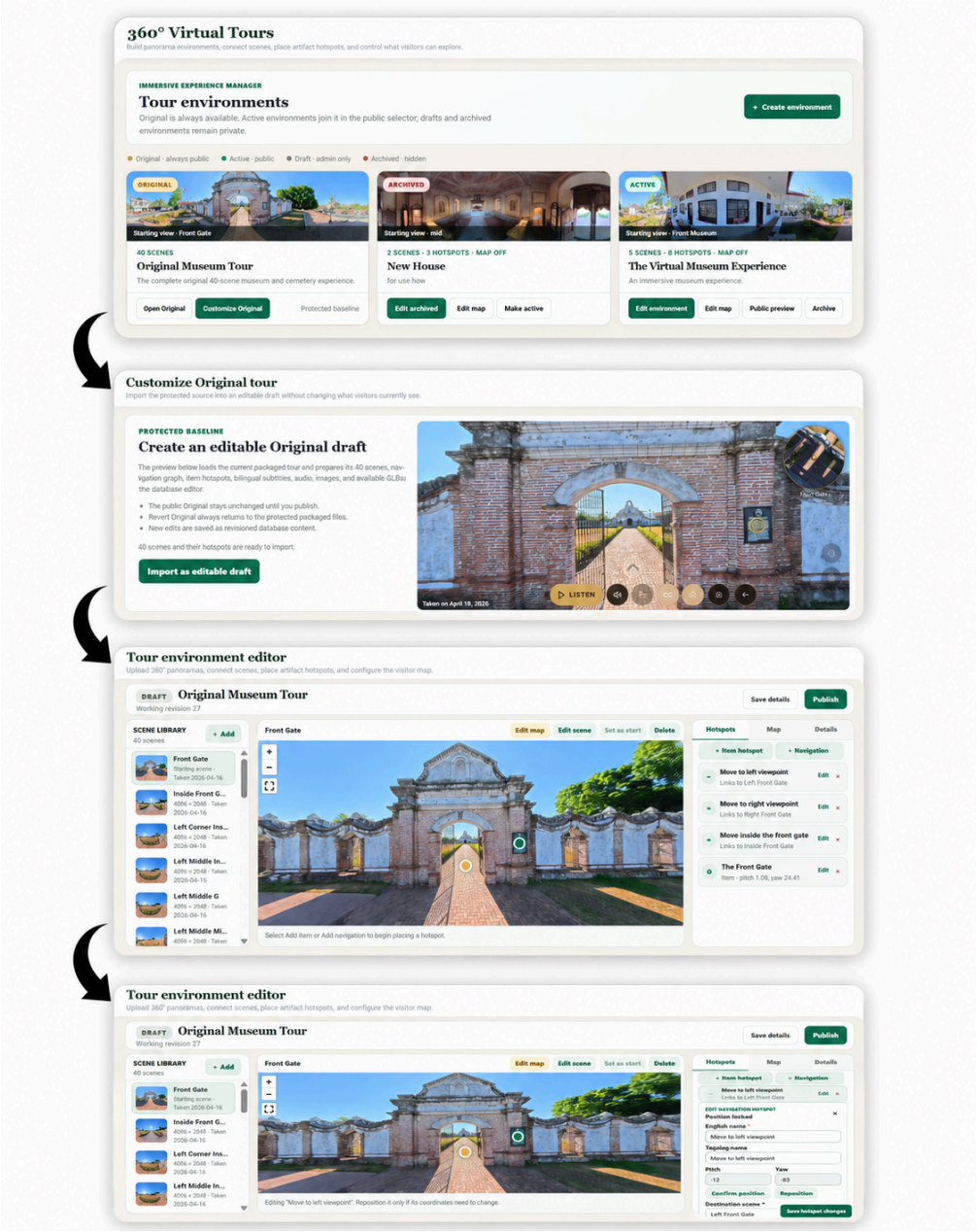

Figure A18. Virtual Tour Environment, Scene, and Hotspot Management

Figure A18 shows how administrators maintain immersive tour content using the Tour Environments library and Virtual Tour editor. Authorized personnel can create a New Environment or use the Original-import workflow, organize scene panoramas and scene settings, create or edit hotspots, preserve active, draft, or archived tours, and make the published environment available in the public tour selector.

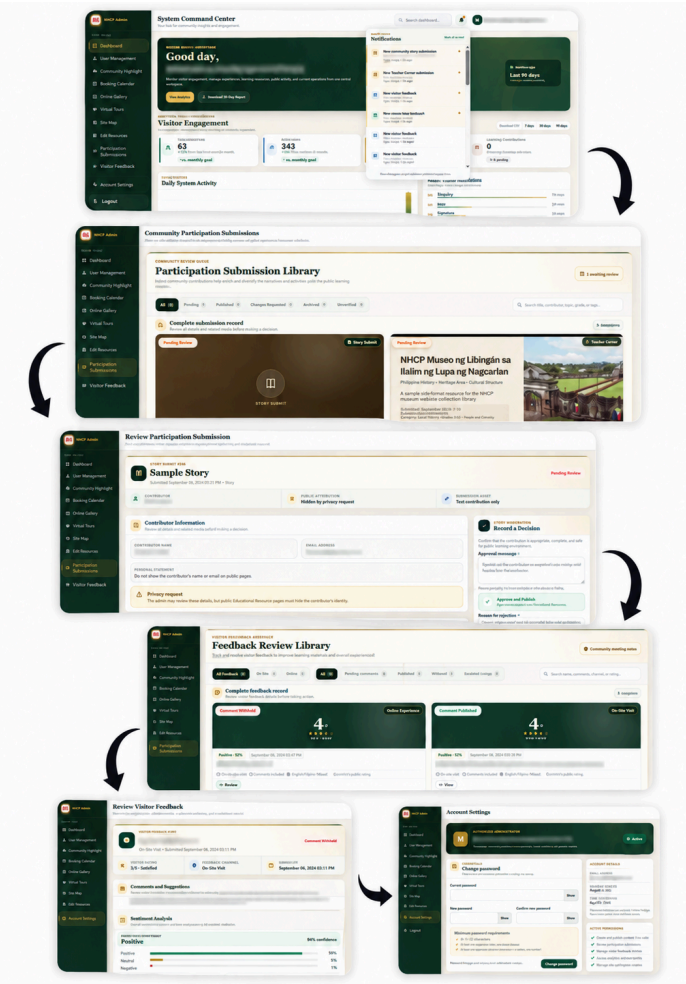

Figure A19. Guided-Visit Booking Review, Participation-Submission Moderation, Visitor-Feedback Review, Notifications, and Account Security

Figure A19 presents the administrator workflows for Guided-Visit Booking Review, Participation-Submission Moderation, Visitor-Feedback Review, Notifications, and Account Security. Administrators can use Booking Operations to review requests and issue Confirm, Reschedule, or Cancel decisions; use the Participation Submission Library to Publish or Archive submissions; use Feedback Analytics and the Feedback Review Library to classify responses; and manage notifications, Administrator Account Settings, password changes, and Sign Out All Sessions.